

Aquifers

Problem 1: Storativity and Specific Storage in a Confined Aquifer

A confined aquifer consists of dense, sandy gravel with a thickness of 100 m and a porosity of 20%. The aquifer is underlain and overlain by low-permeability confining beds.

Given data:

- Aquifer thickness: $b = 100$ m
- Porosity: $n = 0.20$
- Water density: $\rho_w = 1000$ kg/m³
- Gravitational acceleration: $g = 9.81$ m/s²
- Water compressibility: $\beta_w = 4.8 \times 10^{-10}$ m²/N
- Matrix compressibility (dense, sandy gravel): $\beta_p = 5.2 \times 10^{-9}$ to 1.0×10^{-8} m²/N

Tasks:

- Calculate the specific storage due to water compressibility (let's call that S_s^w).
- Calculate the specific storage due to matrix compressibility (let's call that S_s^M) using the range of values given.
- Calculate the total specific storage (S_s) and the storativity (S) of the aquifer.
- If the hydraulic head drops by 100 m over an area of 1×10^9 m², calculate the volume of water released from storage.
- Explain which component (water compressibility or matrix compressibility) contributes more to the total storage, and discuss the physical meaning of this result.

Problem 2: Specific Yield, Retention, and Porosity from a Drainage Experiment

A soil sample was collected and subjected to a drainage experiment to determine its aquifer properties. The following measurements were obtained:

Given data:

- Mass of sample after draining by gravity: 85 g
- Mass of oven-dried sample: 80 g
- Mass of water that drained by gravity: 20 g
- Bulk density of the wet soil: $\rho_b = 1.65 \text{ g/cm}^3$
- Density of water: $\rho_w = 1.0 \text{ g/cm}^3$

Tasks:

- Calculate the total volume of the soil sample (V_T).
- Calculate the volume of water retained in the sample after drainage (V_r).
- Calculate the volume of water that drained by gravity (V_d).
- Calculate the specific retention (S_r), specific yield (S_y), and porosity (n) of the sample.
- Verify that the relationship $n = S_y + S_r$ holds for your calculated values.
- Based on your results, classify this material according to typical aquifer properties and explain what this means for its potential as a water-bearing formation.

Problem 3: Basin-Fill vs Fluvial Aquifer Systems

Compare and contrast basin-fill aquifers with fluvial aquifers, using specific examples from the lecture material.

Tasks:

a) **Basin-Fill Aquifers:**

- Describe the formation process of basin-fill aquifers, using the Basin and Range province as an example.

- Explain how the geometry of these aquifers differs from other types.
- Discuss the relationship between mountain ranges and basin-fill aquifer formation.

b) **Fluvial Aquifers:**

- Distinguish between braided and meandering river systems in terms of their aquifer-forming potential.
- Explain why braided river systems typically create more productive aquifers than meandering systems.
- Describe the sedimentological characteristics that make fluvial deposits good aquifers.

c) **Comparison:**

- Compare the hydraulic properties and heterogeneity of basin-fill versus fluvial aquifers.
- Discuss which type would be more suitable for large-scale groundwater development and why.
- Explain how the depositional environment controls the long-term sustainability of these aquifer systems.

Problem 4: Major Alluvial Aquifer Systems

Compare three major alluvial aquifer systems discussed in the lecture, focusing on their geological characteristics and importance.

Tasks:

a) **High Plains Aquifer System:**

- Describe the geographic extent and geological composition of this aquifer system.
- (Bonus Question) Explain the depositional history of the Ogallala Group and its relationship to the Rocky Mountains.
- Discuss why this aquifer is considered the largest in the United States and its current usage patterns.

b) **Central Valley Aquifer System:**

- Describe the unique surface water system of the Central Valley and its relationship to the Sierra Nevada.
- Explain how alluvial processes from the Sierra Nevada Mountains shape the aquifer characteristics.

- Discuss the variability in grain texture and hydraulic properties across different parts of the valley.

c) **Indo-Gangetic Basin (IGB) Aquifer:**

- Describe the geographic extent and importance of this aquifer system.
- Discuss why this aquifer is considered the world's most important and the challenges it faces.

d) **Comparative Analysis:**

- Compare the depositional environments and sedimentological characteristics of these three systems.
- Discuss the factors that make each system particularly important for groundwater resources.
- Explain how geological history and current tectonic activity influence the sustainability of these aquifer systems.